

Mritunjoy Halder

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EDUCATION

Indian Institute of Science, Bangalore <i>PhD in Robotics and Autonomous Systems</i>	Bengaluru, Karnataka 2026 – Present
Indian Institute of Engineering Science and Technology <i>B.Tech in Information Technology (CGPA 9.1)</i>	Shibpur, West Bengal 2019 – 2023
Jadavpur Vidyapith <i>Senior Secondary Examination (Score 90.2%)</i>	Kolkata, West Bengal 2017 – 2019

EXPERIENCE

Researcher <i>Tata Research Development And Design Centre</i> • Gen3D Researcher specializing in 3D content generation in XR environment from 3D sketch. • Integrated generative modeling and 3D vision to develop simulators for AR/VR-driven robot training. • Designed a fast 3D data generation algorithm. Reduced generation time from over 1 hour to 30 seconds. • Proposed a 3D object retrieval method from sketches. Achieved 60% accuracy, where SOTA benchmark is 49%. • Enabled generation of 3D environments and objects from text or audio prompts for commercial use.	July 2023 – December 2025 Kolkata, India
Research Intern <i>TCS Research Labs</i> • Proposed a spatio-temporal CNN for anomaly detection in surveillance robots. • Achieved 91% accuracy on a benchmark dataset and validated on a real-world robot. • Granted patent and accepted for oral presentation at IEEE IJCNN.	March 2022 – May 2022 Remote
Research Intern <i>NIT Agartala</i> • Designed a hybrid model for real-time image defogging. • Achieved 92% SSIM and 0.1s processing time with fog removal model.	June 2021 – May 2022 Remote
Research Intern <i>NIT Hamirpur</i> • Developed a transformer-based system for sex, accent, and emotion recognition from speech. • Achieved 94.62% (sex), 97.37% (accent), 99.84% (emotion) accuracy.	January 2022 – March 2022 Remote

PUBLICATIONS¹ - RECENT

UltraNormal: Scaling Diffusion-based Surface Normal Estimators to Ultra-High-Resolution <i>Wiley The Computer Graphics Forum</i> • Satyam Bhardwaj, Lokender Tiwari, Shivam Ashok Shukla, Mritunjoy Halder, Aalok Gangopadhyay, Brojeshwar Bhowmick	2026
GenieMotion: Learning Object-Centric 4D Motion Representations from a Single Image <i>IEEE/CVF Computer Vision and Pattern Recognition Workshop (CVPRW)</i> • Mritunjoy Halder, Ravi Prakash	2026
Sketch3R: Rapid and Realistic 3D VR Sketch Creation to Shape Retrieval <i>IEEE/CVF Winter Conference on Computer Vision (WACV)</i> • Mritunjoy Halder, Shivam Ashok Shukla, Raghav Mittal, Lokender Tiwari, Brojeshwar Bhowmick	2026
FAV3R-Fast and Accurate 3D VR Sketch to 3D Shape Retrieval <i>IEEE International Symposium on Mixed and Augmented Reality (ISMAR)</i> • Mritunjoy Halder*, Shivam Ashok Shukla*, Lokender Tiwari, Raghav Mittal, Brojeshwar Bhowmick	2025
InGenCo-Integrated In-Place 3D Scenario Generation and Collaboration <i>IEEE International Symposium on Mixed and Augmented Reality (ISMAR)</i> • Raghav Mittal, Lokender Tiwari, Satyam Bhardwaj, Mritunjoy Halder, Brojeshwar Bhowmick	2025
GeMR-Multimodal Interactive 3D Scene CComposition in XR <i>ACM Indian Conference on Computer Vision, Graphics and Image Processing(Best Paper Award)</i> • Raghav Mittal, Lokender Tiwari, Shivam Ashok Shukla, Mritunjoy Halder, Brojeshwar Bhowmick	2025
A Transmission Model based Deep Neural Network for Image Dehazing <i>Multimedia Tools and Applications, Springer</i> • Tannistha Pal*, Mritunjoy Halder*, Sattwik Barua*	2024
Anomalous Activity Detection from Ego View Camera of Surveillance Robots <i>IEEE International Joint Conference on Neural Networks (IJCNN)</i> • Mritunjoy Halder, Snehasis Banerjee, Balamuralidhar P	2023

¹For full list visit [Google Scholar](#)

PATENTS

Anomalous Activity Detection for Mobile Surveillance Robots <i>US Patent</i> Inventors: Mritunjoy Halder, Snehasis Banerjee, Balamuralidhar P	Granted
Real-Time Dynamic Scene Dehazing using Deep Residual Neural Networks <i>Indian Patent</i> Inventors: Tannistha Pal, Mritunjoy Halder, Sattwik Barua	Published
FAV3R — Fast and Accurate 3D VR Sketch to 3D Shape Retrieval <i>Patent Application</i> Inventors: Mritunjoy Halder, Shivam Ashok Shukla, Lokender Tiwari, Raghav Mittal, Brojeshwar Bhowmick	Filed
GeMR—Multimodal Interactive 3D Scene COmposition in XR <i>Patent Application</i> Inventors: Raghav Mittal, Lokender Tiwari, Shivam Ashok Shukla, Mritunjoy Halder, Brojeshwar Bhowmick	Filed
UltraNormal: Scaling Diffusion-based Surface Normal Estimators to Ultra-High-Resolution <i>Patent Application</i> Inventors: Satyam Bhardwaj, Lokender Tiwari, Shivam Ashok Shukla, Mritunjoy Halder, Aalok Gangopadhyay, Brojeshwar Bhowmick	Filed

WORK IN PROGRESS

Training-Free Object-Centric 4D Motion from a Single Image <i>Preprint in Preparation</i> Mritunjoy Halder, Ravi Prakash	2026
Simultaneous Trajectory Estimation and Planning(STEAP) in Extreme Environments <i>Work In Progress</i> Mritunjoy Halder, Ravi Prakash	2026

PROJECTS

Improved Diagnosis on Low Resolution Medical Images (B.Tech Thesis) <i>Dual GAN, Python</i> - Proposed a dual GAN framework to enhance low-res medical images for better diagnosis. - Introduced a novel loss function for improved ROI reconstruction. - Achieved 91% accuracy, outperforming SOTA.
Cartoon Emotion Recognition (Won 1st Place) <i>Transformer, Python, PAN India Hackathon</i> - Developed a transformer-based model for sentiment classification using a custom cartoon dataset. - Trained using synthetic images; achieved 95% accuracy on Hackathon dataset.

PROFESSIONAL SERVICES

Reviewer <i>IEEE International Joint Conference on Neural Networks (IJCNN)</i> Served as reviewer for Technical Program Committee (TPC).
TPC Reviewer <i>IEEE International Joint Conference on Artificial Intelligence (IJCAI)</i> Served as reviewer.
Reviewer <i>Conference On Robot Learning (CoRL)</i> Served as reviewer.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, MATLAB, LaTeX
- **Technologies:** Blender (BPY), GANs, CNNs, Transformers, Pytorch, 3D Libraries.
- **Domains:** GenAI (VLM/LLM/3D), Computer Vision/Graphics, Robotics, Control, Imitation Learning, Deep Learning